Topic -5 (Part-2): Monte Carlo Methods (Off-Policy)

AIMLCZG512-Deep Reinforcement Learning

Lecture Notes

1. Topics Covered

1. On-policy and off-policy learning, target policy, behaviour policy, and coverage.
2. Trajectory probabilities and importance-sampling ratios for off-policy Monte Carlo learning.
3. Ordinary and weighted importance-sampling estimators for prediction.
4. Bias, variance, and infinite variance in off-policy Monte Carlo estimation.
5. First-visit and every-visit off-policy Monte Carlo estimation.
6. Incremental implementation of weighted importance-sampling estimates.
7. Off-policy Monte Carlo control and the role of greedy target policies.
8. Discounting-aware and per-decision importance sampling as advanced ideas.
9. Application sketches, review questions, and exam-oriented numerical practice.

2. Why Off-Policy Monte Carlo?

In **on-policy** Monte Carlo control, the same policy generates the episodes and is also the policy being evaluated or improved. This is simple and natural, but it creates a tension: the agent must explore enough to learn, while the policy being learned should move toward better actions.

Off-policy learning. In off-policy learning, the learner uses episodes generated by a **behaviour policy** b to learn about a different **target policy** π. The behaviour policy collects experience. The target policy is the policy whose value or optimality we want to estimate.

The main reason for off-policy learning is the exploration–exploitation dilemma. A behaviour policy can remain exploratory and safe for data collection, while the target policy can be greedy or near-greedy. This is also useful when the data already exist: for example, historical logs from an old recommender, past clinical decisions, or episodes generated by a simulator.

2.1. Target Policy, Behaviour Policy, and Coverage

Let $\pi(a \mid s)$ be the policy to be evaluated and let $b(a \mid s)$ be the policy that generates the data. For off-policy prediction to make sense, the behaviour policy must give nonzero probability to every action that the target policy may take:

$$\pi(a \mid s) > 0 \;\Rightarrow\; b(a \mid s) > 0. \tag{1}$$

This is the **coverage assumption**. If π can choose an action that b never chooses in the same state, then no episode generated by b can give evidence about that target-policy action.

Why useful. Off-policy learning allows us to ask: “Using data collected under one policy, what can we say about another policy?” This is valuable for learning from logs, safe exploration, simulation data, and policies that we cannot yet deploy.

Why difficult. A return observed under b is not automatically a return under π. We must correct for the mismatch between the two policies. Importance sampling gives this correction, but it can introduce high variance.

3. Trajectories and Importance Sampling

3.1. Importance Sampling in General

Importance sampling is a general statistical idea for estimating an expectation under one distribution using samples from another distribution. If samples are drawn from b but the desired expectation is under π, then

$$\mathbb{E}_\pi[X] = \mathbb{E}_b\left[\frac{\Pr_\pi(X)}{\Pr_b(X)}X\right], \tag{2}$$

provided the behaviour distribution covers the target distribution. The ratio tells us how much more or less likely the sample is under the target distribution than under the behaviour distribution.

In off-policy Monte Carlo, the "sample" is not a single number. It is a **trajectory**. Therefore the correction ratio becomes a product of policy-probability ratios along the trajectory.

3.2. Trajectory Probability and Ratio Cancellation

A trajectory segment after time t contains actions and states such as

$$A_t, S_{t+1}, A_{t+1}, \ldots, S_T.$$

If the policy is π, the probability of this trajectory segment, conditioned on S_t, is

$$\Pr\{A_t, S_{t+1}, A_{t+1}, \ldots, S_T \mid S_t, A_{t:T-1} \sim \pi\} = \prod_{k=t}^{T-1} \pi(A_k \mid S_k)\, p(S_{k+1} \mid S_k, A_k). \tag{3}$$

If the same trajectory is generated by the behaviour policy b, the corresponding probability is

$$\Pr\{A_t, S_{t+1}, A_{t+1}, \ldots, S_T \mid S_t, A_{t:T-1} \sim b\} = \prod_{k=t}^{T-1} b(A_k \mid S_k)\, p(S_{k+1} \mid S_k, A_k). \tag{4}$$

Therefore the relative probability of the trajectory under π and under b is

$$\rho_{t:T-1} = \frac{\prod_{k=t}^{T-1} \pi(A_k \mid S_k) p(S_{k+1} \mid S_k, A_k)}{\prod_{k=t}^{T-1} b(A_k \mid S_k) p(S_{k+1} \mid S_k, A_k)} = \prod_{k=t}^{T-1} \frac{\pi(A_k \mid S_k)}{b(A_k \mid S_k)}. \tag{5}$$

Role of model dynamics. The transition probabilities $p(S_{k+1} \mid S_k, A_k)$ cancel from the ratio. Thus the importance-sampling correction needs action probabilities under π and b, not the environment transition model. This is why off-policy Monte Carlo can still be model-free.

3.3. Why We Turn to Importance Sampling

If an episode is generated by b, its return G_t is a sample from the behaviour distribution, not from the target distribution. Importance sampling transforms it into a target-policy estimate:

$$\mathbb{E}_b\left[\rho_{t:T-1} G_t \mid S_t = s\right] = v_\pi(s). \tag{6}$$

The proof idea is direct: multiply each return by the ratio of its probability under π to its probability under b; when summing over all possible trajectories, the behaviour probability is cancelled and the expectation becomes the target-policy expectation.

3.4. A Numerical Trajectory Probability

Suppose a trajectory has three decisions:

$$(s_0, a_1, s_1, a_2, s_0, a_1, T).$$

Let

$$\pi(a_1 \mid s_0) = 0.8, \quad b(a_1 \mid s_0) = 0.5, \quad \pi(a_2 \mid s_1) = 0.7, \quad b(a_2 \mid s_1) = 0.4.$$

Then

$$\rho = \frac{0.8}{0.5} \times \frac{0.7}{0.4} \times \frac{0.8}{0.5} = 4.48. \tag{7}$$

If the observed return is $G = 3$, then the ordinary importance-weighted return is $\rho G = 13.44$.

Interpretation. If $\rho > 1$, the trajectory is more likely under the target policy than under the behaviour policy. If $\rho < 1$, the trajectory is less likely under the target policy. If any action has $\pi(A_k \mid S_k) = 0$, the whole ratio becomes zero.

4. Off-Policy Prediction as an Estimation Problem

Consider estimating $v_\pi(s)$ from episodes generated by b. Let G_t be the return after time t, and let $\rho_{t:T(t)-1}$ be the importance-sampling ratio from time t until the termination time $T(t)$ of that episode.

4.1. Clarifying $\mathcal{T}(s)$: Every Visit and First Visit

In the textbook notation, $\mathcal{T}(s)$ is commonly used for the set of time indices at which state s is visited. Used this way, it represents the **every-visit** collection. For **first-visit** estimation, we restrict this set to only the first occurrence of s in each episode.

Notation in this note	Meaning
$\mathcal{T}_{\text{EV}}(s)$	All time steps in all episodes at which $S_t = s$. This gives every-visit off-policy MC.
$\mathcal{T}_{\text{FV}}(s)$	Only the first time step in each episode at which $S_t = s$. This gives first-visit off-policy MC.

For example, if one episode has states A, B, A, T, then $\mathcal{T}_{\text{EV}}(A)$ receives two indices from that episode, while $\mathcal{T}_{\text{FV}}(A)$ receives only the first one. The same distinction is used later in the algorithms: first-visit algorithms check whether the state or state-action pair appeared earlier in the episode.

4.2. Ordinary and Weighted Estimators

Using a generic visit set $\mathcal{T}(s)$, the ordinary importance-sampling estimator is

$$V_{\text{OIS}}(s) = \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|}. \tag{8}$$

The weighted importance-sampling estimator is

$$V_{\text{WIS}}(s) = \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}}. \tag{9}$$

Estimator	More detailed character
Ordinary IS	A simple average of importance-weighted returns. For first-visit MC it is unbiased, because each first-visit return is corrected into a target-policy return. Its weakness is variance: a few large ratios can dominate the average.
Weighted IS	A normalized weighted average. It is biased in finite samples because the denominator is random, but the bias decreases asymptotically. In practice it is often more stable because the estimate remains within the range of the sampled returns when returns are bounded.

The detailed bias and variance discussion comes in Section 5. At this stage, remember the operational difference: ordinary IS divides by number of visits; weighted IS divides by total importance weight.

4.3. Example 1: Long-Trajectory Weighting Can Dominate

Let the states be X and Y, and the actions be L, M, R. Assume $\gamma = 0.9$. The target and behaviour policies are:

State	$\pi(L)$	$\pi(M)$	$\pi(R)$	$b(L)$	$b(M)$	$b(R)$
X	0.6	0.3	0.1	0.3	0.4	0.3
Y	0.2	0.5	0.3	0.4	0.3	0.3

Two episodes starting from X are observed under b.

Episode 1. $X \xrightarrow{L,1} Y \xrightarrow{M,0} X \xrightarrow{L,2} Y \xrightarrow{R,1} X \xrightarrow{M,0} Y \xrightarrow{M,4} T$.

$$G_0^{(1)} = 1 + 0.9(0) + 0.9^2(2) + 0.9^3(1) + 0.9^4(0) + 0.9^5(4) = 5.71096, \tag{10}$$

$$\rho^{(1)} = \frac{0.6}{0.3}\frac{0.5}{0.3}\frac{0.6}{0.3}\frac{0.3}{0.3}\frac{0.3}{0.4}\frac{0.5}{0.3} = 8.3333. \tag{11}$$

Episode 2. $X \xrightarrow{R,0} Y \xrightarrow{L,2} X \xrightarrow{M,-1} Y \xrightarrow{M,0} X \xrightarrow{L,1} Y \xrightarrow{R,5} T$.

$$G_0^{(2)} = 0 + 0.9(2) + 0.9^2(-1) + 0.9^3(0) + 0.9^4(1) + 0.9^5(5) = 4.59855, \tag{12}$$

$$\rho^{(2)} = \frac{0.1}{0.3}\frac{0.2}{0.4}\frac{0.3}{0.4}\frac{0.5}{0.3}\frac{0.6}{0.3}\frac{0.3}{0.3} = 0.4167. \tag{13}$$

Thus

$$V_{\text{OIS}}(X) = \frac{8.3333(5.71096) + 0.4167(4.59855)}{2} = 24.7537, \tag{14}$$

whereas

$$V_{\text{WIS}}(X) = \frac{8.3333(5.71096) + 0.4167(4.59855)}{8.3333 + 0.4167} = 5.6580. \tag{15}$$

Reading the result. Ordinary IS is pulled upward because one trajectory receives a large weight. Weighted IS normalizes by the total weight, so it behaves like a weighted average of the observed returns.

4.4. Example 2: Short Trajectory Calculation

Use two states X, Y and two actions L, R, with $\gamma = 0.9$. Let

$$\pi(L \mid X) = 0.8, \quad b(L \mid X) = 0.5, \quad \pi(R \mid Y) = 0.7, \quad b(R \mid Y) = 0.4.$$

The complementary probabilities are used for the other actions. The two observed episodes are:

	Trajectory	Return	Ratio
E_1	$X \xrightarrow{L,0} Y \xrightarrow{R,2} X \xrightarrow{L,1} T$	2.61	4.48
E_2	$X \xrightarrow{R,0} Y \xrightarrow{L,2} X \xrightarrow{R,0} T$	1.80	0.08

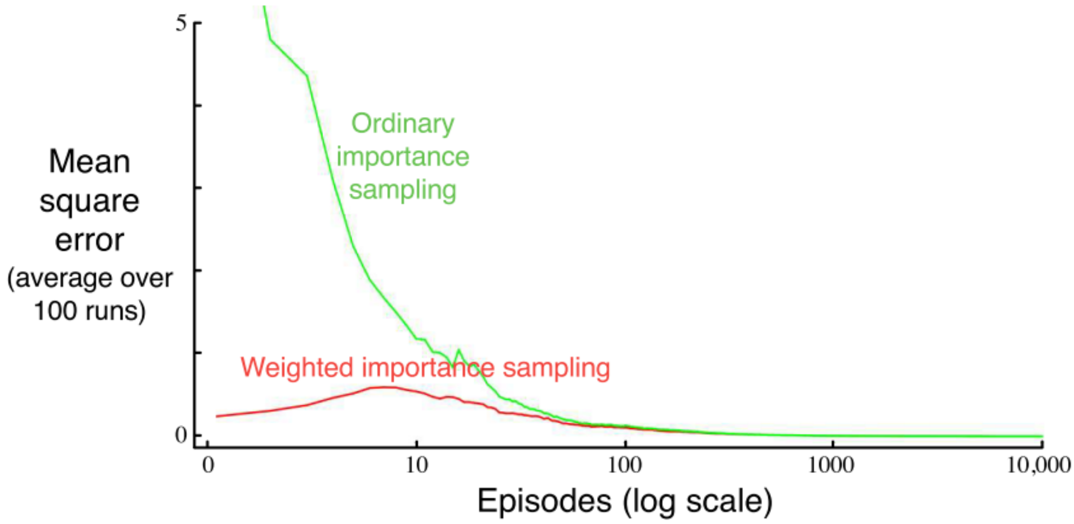
The estimates are

$$V_{\text{OIS}}(X) = \frac{4.48(2.61) + 0.08(1.80)}{2} = 5.9184, \tag{16}$$

$$V_{\text{WIS}}(X) = \frac{4.48(2.61) + 0.08(1.80)}{4.48 + 0.08} = 2.5958. \tag{17}$$

5. Bias, Variance, and Infinite Variance

Importance sampling is powerful because it estimates expectations under one policy using samples from another policy. The difficulty is that products of probability ratios can become very large, especially for long episodes.



Ordinary and weighted importance sampling in the blackjack example.

5.1. How to Read the Blackjack Figure

The figure compares the mean-square error of ordinary and weighted importance sampling in an off-policy blackjack state-value estimation problem. The horizontal axis is the number of episodes on a log scale. The vertical axis is the mean-square error averaged over repeated runs. The green curve corresponds to ordinary importance sampling; it begins with large error and can fluctuate strongly because a small number of high-ratio episodes can dominate the estimate. The red curve corresponds to weighted importance sampling; it is biased in early finite samples, but it is much more stable and has much lower error in the early part of learning.

Inference from the figure. In off-policy Monte Carlo prediction, ordinary IS is theoretically attractive because of unbiasedness in the first-visit case, but weighted IS is often preferred in practice because it controls variance much better.

5.2. Ordinary vs Weighted Importance Sampling

Property	Ordinary importance sampling	Weighted importance sampling
Finite-sample bias	Unbiased for first-visit MC estimates; every-visit variants are generally biased but consistent.	Biased in finite samples because of random normalization.
Variance	Can be very large, and may be infinite.	Usually much smaller; bounded by return range when returns are bounded.
Asymptotic behaviour	Converges under suitable conditions, but may be unstable in practice.	Bias goes to zero asymptotically; often preferred in practice.

Simple interpretation. Ordinary IS asks: "what is the average of importance-weighted returns?" Weighted IS asks: "after giving more weight to target-like trajectories, what is the normalized weighted average?"

5.3. Why Variance Can Explode

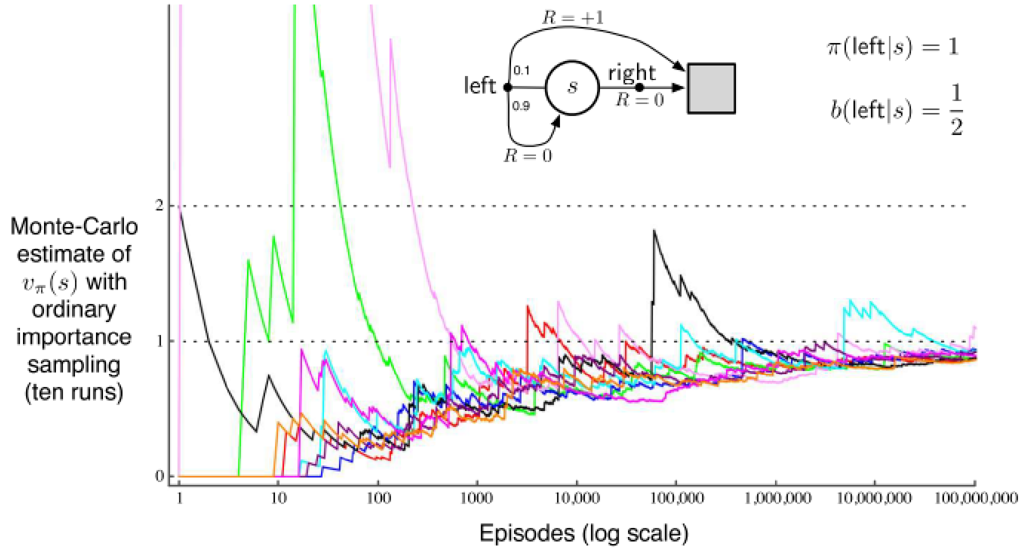
The importance-sampling ratio is a product:

$$\rho_{0:T-1} = \frac{\pi(A_0 \mid S_0)}{b(A_0 \mid S_0)} \frac{\pi(A_1 \mid S_1)}{b(A_1 \mid S_1)} \cdots \frac{\pi(A_{T-1} \mid S_{T-1})}{b(A_{T-1} \mid S_{T-1})}. \tag{18}$$

Even moderate ratios can become large after multiplication over many steps. This is the central reason off-policy Monte Carlo prediction can have high variance.

5.4. Infinite Variance Example

The textbook's one-state example shows the issue sharply. There is one nonterminal state s and two actions: left and right. The target policy always chooses left, while the behaviour policy chooses left and right equally.



The one-state example: ordinary importance sampling can remain unstable for a very large number of episodes.

Under the target policy, the value is $v_\pi(s) = 1$ when $\gamma = 1$, because repeated left actions eventually terminate with reward $+1$. Under the behaviour policy, many episodes choose right and terminate with reward 0; only episodes consisting entirely of left actions until termination contribute to the ordinary IS estimate.

For an episode with n left actions before termination, the importance ratio is 2^n. The second moment contains terms of the form

$$\mathbb{E}\left[(\rho G)^2\right] = 0.1 \sum_{k=0}^{\infty} 0.9^k 2^{k+1} = 0.2 \sum_{k=0}^{\infty} 1.8^k = \infty. \tag{19}$$

Thus the ordinary importance-sampling estimator has infinite variance in this example.

Important point. Infinite variance does not mean the formula is wrong. It means the estimator can be mathematically unbiased but practically unreliable because rare trajectories receive extremely large weights.

6. First-Visit and Every-Visit Off-Policy MC

In off-policy prediction, first-visit and every-visit ideas still apply, but each return is corrected by an importance-sampling ratio.

Variant	How it works in off-policy prediction
First-visit	For each episode, use only the first visit to state s and multiply the following return by the ratio from that time onward.
Every-visit	Use every visit to s in the episode, each with its own suffix ratio $\rho_{t:T-1}$.

6.1. A Small Comparison Example

Consider one episode:

$$S_0 = A, \quad S_1 = B, \quad S_2 = A, \quad S_3 = T.$$

Suppose A occurs twice, with returns $G_0 = 5$ and $G_2 = 2$. Let the suffix ratios be $\rho_{0:2} = 3$ and $\rho_{2:2} = 1.5$.

For state A, first-visit ordinary IS uses only

$$\rho_{0:2} G_0 = 3(5) = 15. \tag{20}$$

Every-visit ordinary IS uses both

$$\rho_{0:2} G_0 = 15, \qquad \rho_{2:2} G_2 = 1.5(2) = 3. \tag{21}$$

With weighted IS, the corresponding first-visit estimate from this one episode is $15/3 = 5$, whereas the every-visit weighted estimate is

$$\frac{15+3}{3+1.5} = 4. \tag{22}$$

> **Important point.** First-visit is cleaner to explain and analyse. Every-visit uses more data from each episode, but multiple returns from the same episode are correlated. In off-policy MC, both variants inherit the variance issue from importance ratios.

7. Incremental Implementation

A batch weighted importance-sampling estimate can be updated incrementally. Suppose we observe returns $G_1, G_2, \ldots$ with corresponding weights $W_1, W_2, \ldots$, where W_i is the appropriate importance-sampling ratio.

The weighted estimate after $n-1$ returns is

$$V_n = \frac{\sum_{k=1}^{n-1} W_k G_k}{\sum_{k=1}^{n-1} W_k}, \qquad n \geq 2. \tag{23}$$

The update can be written as

$$V_{n+1} = V_n + \frac{W_n}{C_n}\left[G_n - V_n\right], \qquad n \geq 1, \tag{24}$$

where C_n is the cumulative sum of weights used in the denominator. In algorithmic form it is often implemented as

$$C \leftarrow C + W, \qquad V \leftarrow V + \frac{W}{C}(G - V). \tag{25}$$

> **Numerical incremental update.** Suppose the weighted returns for a state arrive as $(W, G) = (2,5), (1,2), (3,4)$. Start with $V = 0$ and $C = 0$.

Observation	**New C**	**Update**	**New V**
$(2,5)$	2	$0 + \frac{2}{2}(5-0)$	5
$(1,2)$	3	$5 + \frac{1}{3}(2-5)$	4
$(3,4)$	6	$4 + \frac{3}{6}(4-4)$	4

> The result matches the batch weighted average $\frac{2(5)+1(2)+3(4)}{2+1+3} = 4$.

8. Off-Policy Monte Carlo Control

For prediction, off-policy MC estimates the value of a fixed target policy. For control, the target policy is also improved. Read the loop in simple terms as follows. The behaviour policy b generates a complete episode. The episode is scanned backward. The return G is built step by step. A cumulative weight W tells how consistent the suffix of the episode is with the current target policy. The action-value estimate $Q(S_t, A_t)$ is updated with weighted averaging. Then the target policy is made greedy with respect to the updated Q. If the observed action disagrees with the greedy target action, the earlier part of that episode cannot contribute to the deterministic target policy, so the backward scan stops.

We skip reproducing the off-policy prediction algorithm here. In words, off-policy prediction scans each episode backward, computes G, multiplies by the appropriate importance-sampling weight, and updates the estimated value of the target policy from data generated by the behaviour policy.

8.1. Off-Policy MC Control Algorithm

Off-policy MC control, for estimating $\pi \approx \pi_*$

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:

$$Q(s,a) \in \mathbb{R} \quad \text{arbitrarily}, \qquad C(s,a) = 0,$$
$$\pi(s) \leftarrow \arg\max_a Q(s,a) \quad \textit{initialize target policy greedily; break ties consistently.}$$

Loop forever, for each episode:

1. $b \leftarrow$ any soft policy. *keeps coverage of the greedy target policy.*
2. Generate an episode using b: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$.
3. $G \leftarrow 0$, $W \leftarrow 1$. *return and importance weight are built backward.*
4. Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
   (a) $G \leftarrow \gamma G + R_{t+1}$. *update the return from this time step.*
   (b) $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$. *accumulate total weight.*
   (c) $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \dfrac{W}{C(S_t, A_t)}[G - Q(S_t, A_t)]$. *weighted incremental average.*
   (d) $\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$. *improve target policy greedily.*
   (e) If $A_t \neq \pi(S_t)$, then exit inner loop. *earlier target probability becomes zero.*
   (f) $W \leftarrow W \dfrac{1}{b(A_t \mid S_t)}$. *since target probability is one on the greedy action.*

Why the break condition appears. The target policy is deterministic greedy. If the observed action at time t is not the greedy action under π, then $\pi(A_t \mid S_t) = 0$. The importance ratio for any earlier state-action pair would contain this zero factor, so there is no contribution to earlier updates.

8.2. Example 1: Line World Control

Consider states w, x, y, z with terminal exits from w and z. The action *Exit* at w gives -100; the action *Exit* at z gives $+10$. From x and y, actions are L and R. Let $\gamma = 0.9$. Let the behaviour policy choose L and R with probability 0.5 in x and y.

Episode 1 generated by b.

$$x \xrightarrow{R,0} y \xrightarrow{R,0} z \xrightarrow{Exit,+10} T.$$

Backward updates: At z, $G = 10$, so $Q(z, Exit) = 10$. At y, $G = 0 + 0.9(10) = 9$, so $Q(y, R) = 9$ and $\pi(y) = R$. Since $b(R \mid y) = 0.5$, the next weight becomes $W = 2$. At x, $G = 0 + 0.9(9) = 8.1$, so $Q(x, R) = 8.1$ and $\pi(x) = R$.

Episode 2 generated by b.

$$x \xrightarrow{L,0} w \xrightarrow{Exit,-100} T.$$

At w, $G = -100$, so $Q(w, Exit) = -100$. At x, $G = 0 + 0.9(-100) = -90$, so $Q(x, L) = -90$. Since $Q(x, R) = 8.1 > Q(x, L)$, the greedy target policy still selects R at x. The observed action L disagrees with $\pi(x) = R$, so the inner loop stops for earlier updates.

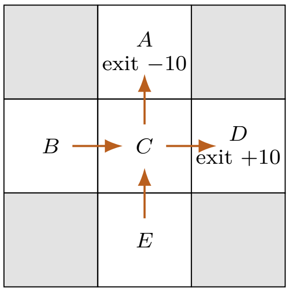
After these two episodes, the target policy is greedy toward the right at x and y:

$$\pi(x) = R, \qquad \pi(y) = R, \qquad \pi(w) = Exit, \qquad \pi(z) = Exit. \tag{26}$$

The behaviour policy may still explore left and right, but the target policy being learned is greedy.

8.3. Example 2: Grid Control

Use the five-state grid from the on-policy note. State D exits with reward $+10$, state A exits with reward -10, and ordinary moves have reward -1. At C, the target policy must learn whether to move toward D or toward A. Let $\gamma = 0.9$ and let b explore both actions at C.



Episode 1.

$$B \xrightarrow{E,-1} C \xrightarrow{E,-1} D \xrightarrow{Exit,+10} T.$$

At D: $Q(D, Exit) = 10$. At C: $G = -1 + 0.9(10) = 8$, so $Q(C, E) = 8$ and $\pi(C) = E$. At B: $G = -1 + 0.9(8) = 6.2$, so $Q(B, E) = 6.2$.

Episode 2.

$$E \xrightarrow{N,-1} C \xrightarrow{N,-1} A \xrightarrow{Exit,-10} T.$$

At A: $Q(A, Exit) = -10$. At C: $G = -1 + 0.9(-10) = -10$, so $Q(C, N) = -10$. Since $Q(C, E) = 8$, the target policy remains $\pi(C) = E$. At this point the observed action N disagrees with the greedy target action at C, so earlier off-policy updates stop.

The target policy has learned to move from C toward D, while the behaviour policy may continue to sample the poorer action N to maintain coverage.

9. Discounting-Aware Importance Sampling

Not covered in class. This section is included only to show where the textbook goes after the standard off-policy MC treatment.

Standard importance sampling applies a single ratio to the whole return. If the episode is long and $\gamma < 1$, very late rewards have little influence on G_t, but their action-probability ratios may still multiply into $\rho_{t:T-1}$. This can add variance without much benefit.

Discounting-aware importance sampling separates the return into parts according to how much future rewards matter. The idea is to avoid multiplying by ratios for far-future actions when those actions affect only heavily discounted rewards. The broad form is to replace the full return by a combination of shorter-horizon partial returns and apply only the ratios needed for those horizons.

Intuition. If a reward is discounted so heavily that it barely affects the return, then action ratios far beyond that reward may add variance without adding useful correction. Discounting-aware methods try to reduce this unnecessary variance.

10. Per-Decision Importance Sampling

Not covered in class. Per-decision importance sampling is a variance-reduction idea. It keeps the same expected target return but corrects each reward term only by the ratios needed up to that reward.

The ordinary off-policy return can be written as

$$\rho_{t:T-1}G_t = \rho_{t:T-1}\left(R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1}R_T\right). \tag{27}$$

Per-decision importance sampling uses the fact that the reward R_{t+k+1} only depends on the trajectory up to that point. Therefore it uses ratios only up to the corresponding decision:

$$\tilde{G}_t = \rho_{t:t}R_{t+1} + \gamma\rho_{t:t+1}R_{t+2} + \cdots + \gamma^{T-t-1}\rho_{t:T-1}R_T. \tag{28}$$

Why it helps. Per-decision IS avoids multiplying early reward terms by later action ratios that could not have affected those earlier rewards. This often reduces variance, particularly in long episodes.

11. Applications of Off-Policy Monte Carlo Ideas

The textbook treatment is tabular and episodic, but the idea of learning about a target policy from data generated by another policy appears widely in applied RL and counterfactual evaluation. The following examples use off-policy correction ideas; practical systems often combine importance sampling with reward models, doubly robust estimators, or function approximation.

11.1. Application 1: Logged Recommendation Systems

Reference. Chen, Beutel, Covington, Jain, Belletti, and Chi, "Top-K Off-Policy Correction for a REINFORCE Recommender System," ACM WSDM 2019. Verified link: `https://arxiv.org/abs/1812.02353`

Why this is off-policy. The behaviour policy is the previous production recommender that generated logged impressions. The target policy is a newer recommendation policy that we want to train or evaluate. Users reveal feedback only for items actually shown by the behaviour policy. Therefore the log is biased toward actions selected by earlier recommenders.

How it is modelled. A state contains user context and session history. An action is a ranked list or top-K recommendation. A reward may be click, watch time, dwell time, or a session-level utility. Off-policy correction reweights logged feedback so that learning is less biased toward the behaviour policy. This is directly related to importance sampling, although production systems adapt the correction for top-K slates and very large action spaces.

11.2. Application 2: Logged Bandit Feedback in Health-Care and Advertising

Reference. Dudik, Langford, and Li, "Doubly Robust Policy Evaluation and Learning," ICML 2011. Verified link: `https://arxiv.org/abs/1103.4601`

Why this is off-policy. Historical decisions are made by clinicians, rules, advertisers, or older decision systems. A new policy cannot be judged by simply averaging observed rewards, because the historical actions do not necessarily match what the new policy would have chosen.

How it is modelled. A context or state describes the patient, user, or request. The action is a treatment choice, ad, or recommendation. The reward is a clinical outcome, click, conversion, or utility. Importance sampling uses the probability with which the historical policy selected the logged action and the probability with which the target policy would select it. The paper also studies doubly robust methods, which combine importance correction with reward modelling to reduce variance.

12. Review of Key Terms

Term	Short definition
Off-policy learning	Learning about a target policy using data generated by another policy.
Target policy π	The policy being evaluated or improved.
Behaviour policy b	The policy used to generate episodes.
Coverage	Requirement that $b(a \mid s) > 0$ whenever $\pi(a \mid s) > 0$.
Importance-sampling ratio	Product of action-probability ratios $\pi(A_k \mid S_k)/b(A_k \mid S_k)$ along a trajectory.
Ordinary importance sampling	Simple average of importance-weighted returns.
Weighted importance sampling	Weighted average normalized by the sum of importance weights.
First-visit off-policy MC	Uses only the first visit to a state or state-action pair in each episode, with importance correction.
Every-visit off-policy MC	Uses every visit, each with its own suffix importance ratio.
Infinite variance	A situation where an estimator has finite expectation but unbounded variance.
Per-decision IS	Importance-sampling correction applied separately to reward terms.
Off-policy MC control	A control method that learns a greedy target policy from episodes generated by a soft behaviour policy.

13. Review Questions

1. Explain the difference between on-policy and off-policy Monte Carlo learning. Identify the role of the target policy and the behaviour policy.
   Hint: On-policy learns about the same policy that acts. Off-policy uses data from b *to evaluate or improve* π*.*
2. State the coverage assumption for off-policy learning. Why is this assumption necessary?
   Hint: Use $\pi(a \mid s) > 0 \Rightarrow b(a \mid s) > 0$*. If* b *never tries an action, no data can support its value under* π*.*
3. For $S_0 = s_0, A_0 = a_0, S_1 = s_1, A_1 = a_1, S_2 = s_2$, write the probability of the trajectory segment under π and under b, including the environment dynamics. Show how the dynamics cancel in the ratio.
   Hint: Write products of policy probabilities and transition probabilities, then divide term by term.
4. Suppose $\pi(a_1 \mid s) = 0.8$, $b(a_1 \mid s) = 0.4$, $\pi(a_2 \mid s') = 0.3$, and $b(a_2 \mid s') = 0.6$. Compute the importance-sampling ratio for the two-step action sequence.
   Hint: $\rho = (0.8/0.4)(0.3/0.6) = 1$*.*
5. Two off-policy returns from state s are $G_1 = 6$ and $G_2 = 2$, with ratios $\rho_1 = 4$ and $\rho_2 = 0.5$. Compute ordinary and weighted importance-sampling estimates.
   Hint: Ordinary: $(4 \cdot 6 + 0.5 \cdot 2)/2 = 12.5$*. Weighted:* $(4 \cdot 6 + 0.5 \cdot 2)/(4 + 0.5) = 25/4.5 \approx 5.56$*.*
6. In Example 1, explain why the ordinary IS estimate is much larger than both observed returns. What does this reveal about variance?
   Hint: The first trajectory has a large ratio. Ordinary IS divides by number of samples, not by total weight.
7. Give a short episode in which first-visit and every-visit off-policy MC use different numbers of samples for the same state.
   Hint: Use A, B, A, T*. First-visit uses the first* A *only; every-visit uses both occurrences with their own suffix ratios.*
8. Explain why weighted importance sampling is biased in finite samples but often preferred in practice.
   Hint: The denominator is random, which creates finite-sample bias. Normalization usually reduces variance.
9. Reproduce the infinite-variance argument for the one-state example. Why does the series diverge?
   Hint: For n *consecutive left actions, the ratio grows like* 2^n*. The second moment contains a geometric series with ratio* $1.8 > 1$*.*
10. In off-policy MC control, why does the algorithm stop the backward scan when $A_t \neq \pi(S_t)$?
    Hint: The target policy is deterministic greedy. If the observed action differs from the target action, the numerator probability becomes zero for earlier suffixes.
11. A behaviour policy chooses actions uniformly among four actions. A deterministic target policy chooses one action in each state. If an episode follows the target action for three consecutive steps, what is the importance ratio for those three steps?
    Hint: Each ratio is $1/(1/4) = 4$*, so the product is* $4^3 = 64$*.*
12. Consider the line-world example. If $x \xrightarrow{R,0} y \xrightarrow{R,0} z \xrightarrow{Exit,+10} T$ and $\gamma = 0.9$, compute the returns for $(z, Exit)$, (y, R), and (x, R).
    Hint: Work backward: 10*,* $0 + 0.9(10) = 9$*, and* $0 + 0.9(9) = 8.1$*.*
13. Explain the difference between ordinary trajectory-level IS and per-decision IS. Why can per-decision IS reduce variance?
    Hint: Trajectory-level IS applies the full ratio to the whole return; per-decision IS applies only the

ratios needed for each reward term.

14. Formulate an off-policy evaluation problem for a recommender system using logged data. Identify states, actions, rewards, behaviour policy, target policy, and why importance correction is needed.
    Hint: Use user context/session as state, recommended items as actions, clicks or watch time as rewards, old recommender as b, new recommender as π.
15. A medical decision system has observational data from doctors but wants to evaluate a new treatment policy. Explain why this is off-policy and identify one risk of using ordinary importance sampling.
    Hint: Doctors' historical decisions are b; the new policy is π. Ordinary IS can have high variance when the target policy differs strongly from observed behaviour.

14. Points to Remember

- Off-policy learning separates the policy used for data generation from the policy being learned.
- Importance sampling corrects for the mismatch between π and b using policy-probability ratios.
- Environment dynamics cancel from the trajectory ratio, so the method can remain model-free.
- Ordinary IS is unbiased for first-visit prediction but can have very high or infinite variance.
- Weighted IS is biased in finite samples but usually has much lower variance.
- Off-policy MC control learns a greedy target policy from data generated by a soft behaviour policy.
- Per-decision and discounting-aware variants are variance-reduction ideas for long episodes.

15. Required Reading

Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press. Required reading: Chapter 5, especially Sections 5.5 to 5.7, covering off-policy prediction via importance sampling, incremental implementation, and off-policy Monte Carlo control. Sections 5.8 and 5.9 may be read as optional advanced material.

16. References

1. Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). Cambridge, MA: MIT Press.
2. Precup, D., Sutton, R. S., and Singh, S. (2000). Eligibility traces for off-policy policy evaluation. *Proceedings of the 17th International Conference on Machine Learning.*
3. Dudik, M., Langford, J., and Li, L. (2011). Doubly robust policy evaluation and learning. *Proceedings of the 28th International Conference on Machine Learning.* `https://arxiv.org/abs/1103.4601`
4. Chen, M., Beutel, A., Covington, P., Jain, S., Belletti, F., and Chi, E. (2019). Top-K off-policy correction for a REINFORCE recommender system. *Proceedings of the ACM International Conference on Web Search and Data Mining.* `https://arxiv.org/abs/1812.02353`
5. Sutton, R. S., Mahmood, A. R., Precup, D., and van Hasselt, H. (2014). A new Q(λ) with interim forward view and Monte Carlo equivalence. *Proceedings of ICML.*

17. Acknowledgements

These lecture notes are based on and adapted from Chapter 5 of Sutton and Barto's *Reinforcement Learning: An Introduction*, Second Edition.

End of Topic -5 (Part-2)